Data Structure and Content Description

In-depth Presentation of the Structure and Content of the Database

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1 - General Definitions and Concepts

What is a Disaster in EM-DAT?

EM-DAT was designed in 1988 based on an anthropocentric vision of disasters and emergencies[1]. It considers disasters to be events involving an unexpected and overwhelming harmful impact on human beings. Formally, EM-DAT's definition of a disaster is:

Definition: **Disaster**

A situation or event which overwhelms local capacity, necessitating a request to the national or international level for external assistance; an unforeseen and often sudden event that causes great damage, destruction, and human suffering.

EM-DAT inventories only disasters that fit its Entry Criteria. EM-DAT records both disasters triggered by natural hazards and technological disasters. The latter are unintentional accidents, and not situations of conflict, violence, or terrorism. For more details, we refer to the Disaster Classification System.

Note

EM-DAT reports disaster impacts per country and per hazard type. However, for some disasters, this simple representation may not capture the situation's complexity, systemic and composite effects, nor its subsequent uncertainties (see General Issues).

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1. Guha-Sapir, D., & Misson, C. (1992). The development of a database on disasters. *Disasters*, *16*(1), 74–80. https://doi.org/10.1111/j.1467-7717.1992.tb00378.x ↩

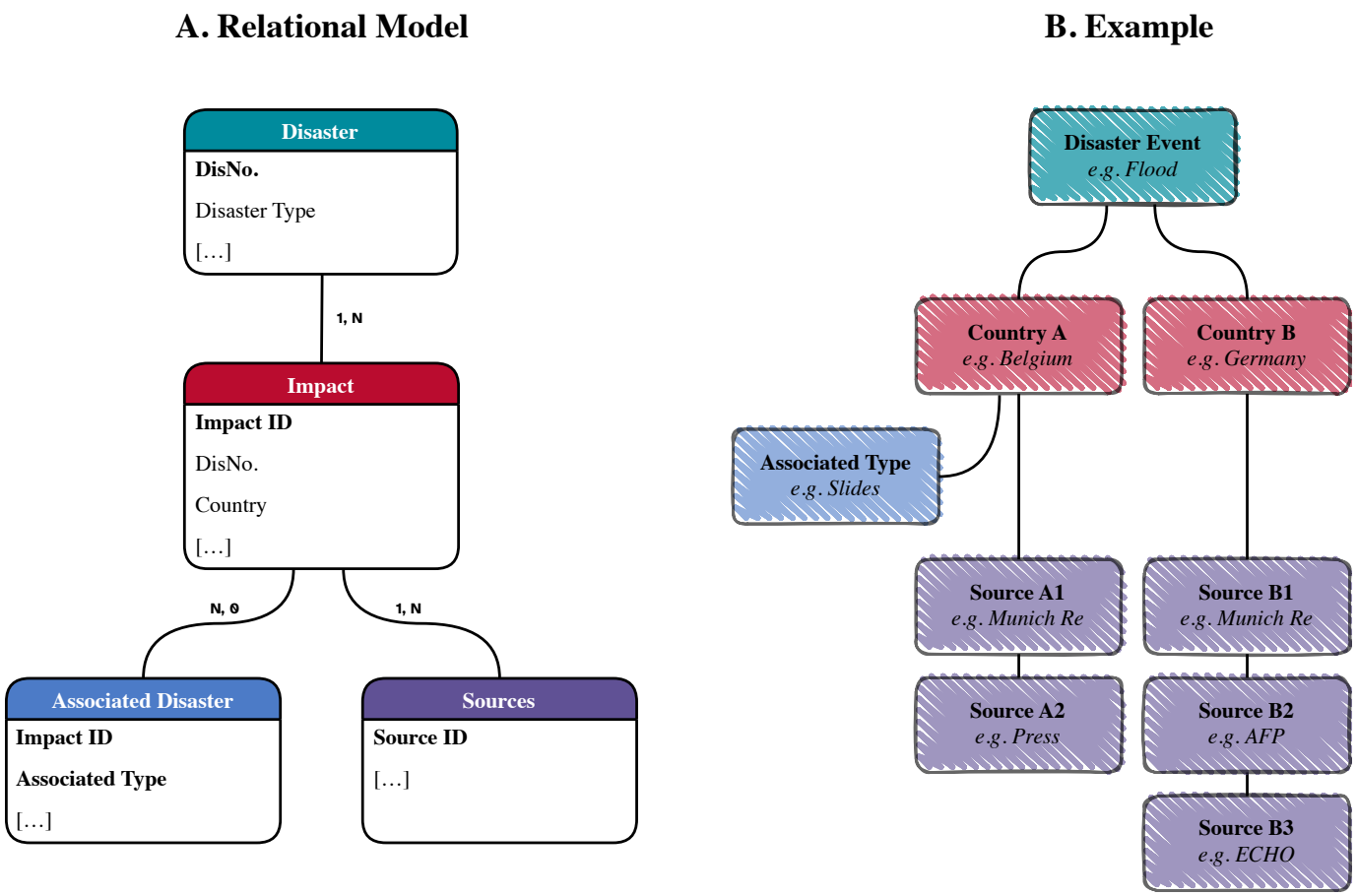
2 - Core Structure of the Database

Basic Relational Structure of the Various Components of a Disaster Event in EM-DAT

The EM-DAT data model records disasters and maps their impacts at the country level. Accordingly, the EM-DAT database references disasters with a unique identifier (see `DisNo.` in Column Description). A disaster may affect more than one country. In EM-DAT, the country-specific information about the disaster is recorded as an “Impact.” The Impact is the level at which EM-DAT reports entries in the EM-DAT Public Table. To document each Impact, we cross-validate and select information from the sources that are available (see EM-DAT Sources and Protocols).

> ### EM-DAT Structure in a Nutshell
>
> - A disaster in EM-DAT may have one or more impacts documented in different countries.
> - Its impact may have one or more sources reporting country-related information.
> - The country-related information, which is selected depending on the trustworthiness of the source, is reported in the public table.

The figure below provides a visual representation of the foundational structure of the EM-DAT database’s relational model. On the left is the core structure, and on the right, a hypothetical case serves as an illustrative example. To enhance clarity, the relations have been streamlined, omitting certain tables such as dictionaries, focusing solely on the main connections and the underlying logic of the database.



EM-DAT Simplified Relational Model with an Example

Cardinality, or the nature of the relationship between entities, is denoted by numbers or the letter “N” on each link. For instance, “1, N” signifies a one-to-many relationship. A clear example of this is the link between the “Disaster” and “Impact” tables. This implies that a single disaster entry might correspond to multiple impacts across various countries. Consider a flood affecting multiple nations, such as Belgium and Germany; the connection between these entries is facilitated by the shared `DisNo.` attribute.

The primary classification of a disaster, such as its main type, is housed in the “Disaster” table. Consequently, all impacts linked by the same `DisNo.` will have a consistent primary disaster type. However, EM-DAT introduces a nuance with associated disasters linked to the “Impact” table. Unlike the main type, these associated disasters can vary by country. They represent secondary disasters that either result from or occur simultaneously with the main event. An example would be a landslide triggered by a primary flood event, which is considered to be an associated disaster. It’s important to note that associated disasters are optional, leading to their zero-to-many relationship with the “Impact” table.

Each source can provide data for any given variable within the Impact category. During the validation process (as detailed in Encoding, Quality Control, and Validation Procedure), one value is chosen for

each variable, which is independent of other variables. Some figures presented in the EM-DAT Public Table are aggregates of various variables and might originate from multiple distinct sources. For example, figures for deaths and missing persons might be reported at different times after an event. After evaluating and selecting from the available data, the `Total Deaths` column might consolidate information from various sources.

3 - EM-DAT Public Table

Content and Presentation of the Database as Publicly Available

Overview

The EM-DAT Public Table is a flat representation of EM-DAT data in a single downloadable table. Most impact variables are part of the public table (see Impact Variables). The public table provides a flat view of the general structure in which each record (row) corresponds to a disaster impacting a country.

About empty cells

For given Impact Figures, empty cells could either refer to a disaster with no impact or a disaster with unknown or unreported impact. See also Accounting Biases.

About deprecated fields

Fields that are not maintained are labeled as **(Deprecated)** in the following table. This usually means that their collection is not possible at the moment or indefinitely stopped. Definitive removal dates are not currently planned. Deprecated fields include:

- **OFDA/BHA Response**
- **AID Contribution ('000 US$)**
- **Admin Units** (users are advised to migrate to the **GADM Admin Units** column)

Column Description

Column Name	Type	Description
DisNo.	ID, Mandatory	A unique 8-digit identifier including the year (4 digits) and a sequential number (4 digits) for each disaster event (e.g., 2004-0659). In the EM-DAT Public Table, the ISO country code is appended. See column **ISO** below.

Column Name	Type	Description
Historic	Yes/No, Mandatory	Binary field specifying whether or not the disaster happened before 2000, using the **Start Year**. Data before 2000 should be considered of lesser quality (see Time Bias).
Classification Key	ID, Mandatory	A unique 15-character string identifying disasters in terms of the **Group**, **Subgroup**, **Type** and **Subtype** classification hierarchy. See Disaster Classification System.
Disaster Group	Name, Mandatory	The disaster group, i.e., “Natural” or “Technological.” See Disaster Classification System.
Disaster Subgroup	Name, Mandatory	The disaster subgroup. See Disaster Classification System.
Disaster Type	Name, Mandatory	The disaster type. See Disaster Classification System.
Disaster Subtype	Name, Mandatory	The disaster subtype. See Disaster Classification System.
External IDs	IDs List, Optional	List of identifiers for external resources (GLIDE, USGS, DFO, HANZE), in the format “<source>:<identifier>” and separated by the pipe character ("|").
Event Name	Text, Optional	Short specification for disaster identification, such as a storm name (e.g., “Mitch”), an aircraft type in air crashes (e.g., “Boeing 707”), a disease name (e.g., “Cholera”), or a volcano name (e.g., “Etna”).
ISO	ID, Mandatory	The International Organization for Standardization (ISO) 3-letter code referring to the **Country**. The ISO 3166 norm is used. See Spatial Information and Geocoding.
Country	Name, Mandatory	Country where the disaster occurred and had an impact, using names from the UN M49 Standard. See Spatial Information and Geocoding. If multiple countries are affected, each will have an entry linked to the same **DisNo.**.
Subregion	Name, Mandatory	Subregion where the disaster occurred based on UN M49 standard, automatically linked to the **Country** field. See Spatial Information and Geocoding.
Region	Name, Mandatory	Region or continent where the disaster occurred based on UN M49 standard, automatically linked to the **Country** field. See Spatial Information and Geocoding.

Column Name	Type	Description
Location	Text, Optional	Geographical location name as specified in the sources, e.g., city, village, department, province, state, or district. Used to identify the corresponding **GADM Admin Units** (see GADM 4.1 Administrative Units).
Origin	Text, Optional	Additional specifications on the contextual factors that led to the event, e.g., “heavy rains” for floods, or “drought” for a forest fire.
Associated Types	Names List, Optional	List of secondary disaster types cascading from or co-occurring with the main type, e.g., a landslide following a flood or an explosion after an earthquake. Separated by the pipe character ("|").
OFDA/BHA Response	Yes/No, Optional	***(Deprecated)*** Binary field specifying whether or not the (former) Office of US Foreign Disaster Assistance (OFDA) or the Bureau for Humanitarian Assistance (BHA) responded to the disaster. Not maintained after 2025.
Appeal	Yes/No, Mandatory	Binary field specifying whether or not there was a request for international assistance from the affected country.
Declaration	Yes/No, Mandatory	Binary field specifying whether a state of emergency was declared in the country.
AID Contribution (‘000 US$)	Unadjusted Monetary Amount (‘000 US$), Optional	***(Deprecated)*** The total amount (in thousands of US$ at the time of the report) of contributions for immediate relief activities to the country in response to the disaster, sourced from the Financial Tracking System of OCHA (1992 to 2015). Not maintained after 2015 due to a lack of availability of information. Some aid contribution information can be found at https://fts.unocha.org/.
Magnitude	Disaster-Type-Dependent, Optional	The intensity of a specific disaster (see Hazard and Disaster Magnitude Units).
Magnitude Scale	Disaster-Type-Dependent, Optional	The associated unit for the **Magnitude** column (see Hazard and Disaster Magnitude Units).
Latitude	Degrees [-90;90], Optional	North-South coordinates mainly for earthquakes and volcanic activity. Sometimes reported for floods, landslides, and storms (mostly when associated with floods).
Longitude	Degrees [-180;180], Optional	East-West coordinates mainly for earthquakes and volcanic activity. Sometimes reported for floods, landslides, and storms (mostly when associated with floods).

Column Name	Type	Description
River Basin	Text, Optional	Name of affected river basins, typically used for floods.
Start Year	Numeric, Mandatory	Year of occurrence of the disaster.
Start Month	Numeric, Optional	Month of occurrence of the disaster. For sudden-impact disasters, this field is well defined. For disasters developing gradually over a longer time period (e.g., drought) with no precise onset date, this field can be left blank.
Start Day	Numeric, Optional	Day of occurrence of the disaster. For sudden-impact disasters, this field is well defined. For disasters developing gradually over a longer time period (e.g., drought) with no precise onset date, this field can be left blank.
End Year	Numeric, Optional	Year of disaster conclusion.
End Month	Numeric, Optional	Month of conclusion of the disaster. For sudden-impact disasters, this field is well defined. For disasters developing gradually over a longer time period (e.g., drought) with no precise end date, this field can be left blank.
End Day	Numeric, Optional	Day of conclusion of the disaster. For sudden-impact disasters, this field is well defined. For disasters developing gradually over a longer time period (e.g., drought) with no precise end date, this field can be left blank.
Total Deaths	Numeric, Optional	Total fatalities (deceased and missing combined, see Human Impact Variables).
No. Injured	Numeric, Optional	Number of people with physical injuries, trauma, or illness requiring immediate medical assistance due to the disaster (see Human Impact Variables).
No. Affected	Numeric, Optional	Number of people requiring immediate assistance due to the disaster (see Human Impact Variables).
No. Homeless	Numeric, Optional	Number of people requiring shelter due to their house being destroyed or heavily damaged during the disaster (see Human Impact Variables).
Total Affected	Numeric, Optional	Total number of affected people (**No. Injured**, **No. Affected**, and **No. Homeless** combined, see Human Impact Variables).

Column Name	Type	Description
Reconstruction Costs (‘000 US$)	Unadjusted Monetary Amount (‘000 US$), Optional	Costs for replacement of lost assets in thousands of US dollars (‘000 US$) relative to **Start Year**, unadjusted for inflation (see Economic Impact Variables).
Reconstruction Costs, Adjusted (‘000 US$)	Adjusted Monetary Amount (‘000 US$), Optional	**Reconstruction Costs (‘000 US$)**, adjusted for inflation using the Consumer Price Index (**CPI** column, see Economic Adjustment).
Insured Damage (‘000 US$)	Unadjusted Monetary Amount (‘000 US$), Optional	Economic damage covered by insurance companies, in thousands of US dollars (‘000 US$), relative to **Start Year**, unadjusted for inflation (see Economic Impact Variables).
Insured Damage, Adjusted (‘000 US$)	Adjusted Monetary Amount (‘000 US$), Optional	**Insured Damage (‘000 US$)** adjusted for inflation using the Consumer Price Index (**CPI** column, see Economic Adjustment).
Total Damage (‘000 US$)	Unadjusted Monetary Amount (‘000 US$), Optional	Value of all economic losses directly or indirectly due to the disaster, in thousands of US dollars (‘000 US$), relative to **Start Year**, unadjusted for inflation (see Economic Impact Variables).
Total Damage, Adjusted (‘000 US$)	Adjusted Monetary Amount (‘000 US$), Optional	**Total Damage (‘000 US$)** adjusted for inflation using the Consumer Price Index (**CPI** column, see Economic Adjustment).
CPI	Conversion Ratio, Optional	Consumer Price Index from OECD used to adjust US$ values for inflation relative to **Start Year** (see Economic Adjustment).
Admin Units	JSON Array of Objects, Optional	***(Deprecated)*** Collection of impacted Administrative Units from the FAO GAUL 2015 referential (Global Administrative Unit Layers 2015). Individual objects correspond to Admin-1 or Admin-2 administrative units, with the corresponding fields `adm1_code`, `adm1_name` or `adm2_code`, `adm2_name` providing the unique identifier to the geometry in the GAUL layer and the name of the unit, respectively. Geocoding is maintained for non-biological natural hazards from 2000 onwards (see Spatial Information and Geocoding). Not maintained after 2025.

Column Name	Type	Description
GADM Admin Units	JSON Array of Objects, Optional	Collection of impacted Administrative Units from the GADM Database of Global Administrative Areas (version 4.1). Individual objects correspond to Admin-1 or Admin-2 administrative units, with a unique identifier (`gid_1` or `gid_2`) mapping to the associated geometries in the GADM referential and the name of the unit (`name_1` or `name_2`). For events before 2026, most GADM units were derived from the GAUL units (**Admin Units** column) using geoprocessing. Migrated objects include additional fields reporting the date (`migration_date`) and specific method (`migration_method`) involved in the process. For more information, see Spatial Information and Geocoding.
Entry Date	Date, Mandatory	The day on which the event record was created in EM-DAT.
Last Update	Date, Mandatory	The last modification of the event or one of its associated records in EM-DAT. This may not result in a modification of the information in the EM-DAT Public Table as modifications to private fields are recorded as well.

4 - Disaster Classification System

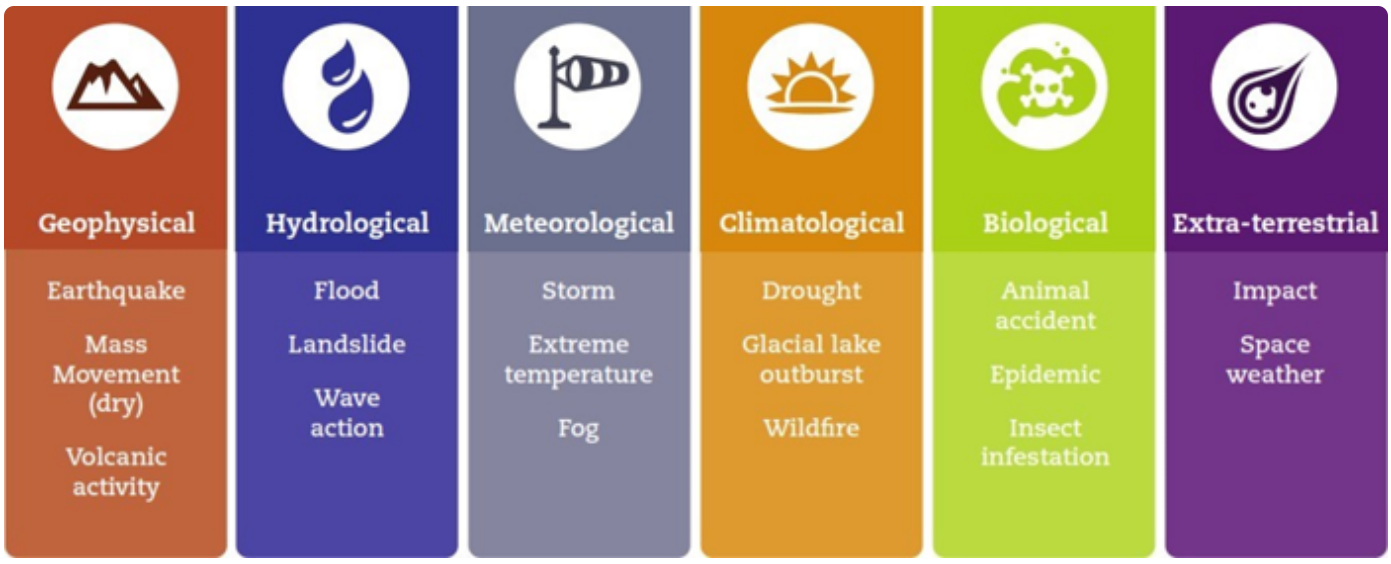
Historical and Current Classification System of Disasters

A Brief History of the EM-DAT Classification System

EM-DAT's classification system originally started with a simple 20-type list[1]. However, in 1992, CRED and other international stakeholders proposed a hierarchical classification system[2] that distinguishes natural and man-made disasters (described as technological disasters in EM-DAT). A further distinction was based on the timing of disasters: slow vs. rapid onset disasters.

In the 2000s, CRED collaborated with Munich Re and other stakeholders on a common classification system[3]. Since then, EM-DAT's main classification system has followed the logic of referring to the hazard or event triggering the disaster. Consequently, some disasters were reclassified, and some types were removed from the primary classification system. This was the case for famines reclassified as drought for the most part[4]. However, "famine" offers more information than "drought." Therefore, in order not to lose this added value, EM-DAT has adopted the secondary classification system of Associated Disasters. This describes disasters that coincide with or result from the primary type.

In 2014, the Integrated Research on Disaster Risk (IRDR) working group, which included CRED, established a new reference called the Peril Classification and Hazard Glossary[5]. This document is currently the primary reference for classifying natural hazards in EM-DAT, which divides them into six main groups: Geophysical, Hydrological, Meteorological, Climatological, Biological, and Extra-terrestrial. EM-DAT also includes more detailed subtypes.



Natural Hazards Subgroups and Types in the IRDR Peril Classification and Hazard Glossary

Note

More recently, in 2021, the ISC and UNDRR released the Hazard Information Profiles (HIPs), updated in 2025. The 2025 edition brings together and defines 281 specific hazards, grouped into eight hazard types, that have the potential to impact a community. In March 2023, CRED discussed HIPs during its Scientific and Technical Advisory Group (STAG) meeting in Brussels. As a result, CRED is working on a roadmap to make its classification system interoperable with HIPs.

Main Classification Tree

Changes in the Classification Tree

In September 2023, the EM-DAT Classification Tree was updated and simplified to facilitate user queries and filtering. The current tree, which is based on four depth levels, replaces the former tree with three plus two optional depth levels. The 'Complex' disaster group (major famine situations for which drought was not the main causal factor) was removed (14 entries). If you wish to establish the correspondence between new and former disaster types, you may download this backup of EM-DAT prior to the update and join the two datasets based on the `DisNo.` column.

The main classification tree has four levels of depth, so disasters are divided into groups, subgroups, types, and subtypes, as presented in the EM-DAT Public Table columns. The two EM-DAT disaster groups are 'Natural' and 'Technological'. The table below shows the complete tree for the 'Natural' and 'Technological' groups, with the occurrence for each subtype. Their corresponding definitions are available in the Classification Glossary.

Classification Key	Group	Subgroup	Type	Subtype	Count[6]
nat-bio-ani-ani	Natural	Biological	Animal incident	Animal incident	1
nat-bio-epi-bac	Natural	Biological	Epidemic	Bacterial disease	792
nat-bio-epi-dis	Natural	Biological	Epidemic	Infectious disease (General)	143
nat-bio-epi-fun	Natural	Biological	Epidemic	Fungal disease	0
nat-bio-epi-par	Natural	Biological	Epidemic	Parasitic disease	52

Classification Key	Group	Subgroup	Type	Subtype	Count[6]
nat-bio-epi-pri	Natural	Biological	Epidemic	Prion disease	0
nat-bio-epi-vir	Natural	Biological	Epidemic	Viral disease	528
nat-bio-inf-gra	Natural	Biological	Infestation	Grasshopper infestation	16
nat-bio-inf-inf	Natural	Biological	Infestation	Infestation (General)	9
nat-bio-inf-loc	Natural	Biological	Infestation	Locust infestation	67
nat-bio-inf-wor	Natural	Biological	Infestation	Worms infestation	3
nat-cli-dro-dro	Natural	Climatological	Drought	Drought	796
nat-cli-glo-glo	Natural	Climatological	Glacial lake outburst flood	Glacial lake outburst flood	5
nat-cli-wil-for	Natural	Climatological	Wildfire	Forest fire	315
nat-cli-wil-lan	Natural	Climatological	Wildfire	Land fire (Brush, Bush, Pasture)	92
nat-cli-wil-wil	Natural	Climatological	Wildfire	Wildfire (General)	81
nat-ext-imp-air	Natural	Extra-terrestrial	Impact	Airburst	0
nat-ext-imp-col	Natural	Extra-terrestrial	Impact	Collision	1
nat-ext-spa-ene	Natural	Extra-terrestrial	Space weather	Energetic particles	0
nat-ext-spa-geo	Natural	Extra-terrestrial	Space weather	Geomagnetic storm	0
nat-ext-spa-rad	Natural	Extra-terrestrial	Space weather	Radio disturbance	0
nat-ext-spa-sho	Natural	Extra-terrestrial	Space weather	Shockwave	0
nat-geo-ear-gro	Natural	Geophysical	Earthquake	Ground movement	1563
nat-geo-ear-tsu	Natural	Geophysical	Earthquake	Tsunami	56
nat-geo-mmd-ava	Natural	Geophysical	Mass movement (dry)	Avalanche (dry)	5
nat-geo-mmd-lan	Natural	Geophysical	Mass movement (dry)	Landslide (dry)	30
nat-geo-mmd-roc	Natural	Geophysical	Mass movement (dry)	Rockfall (dry)	10

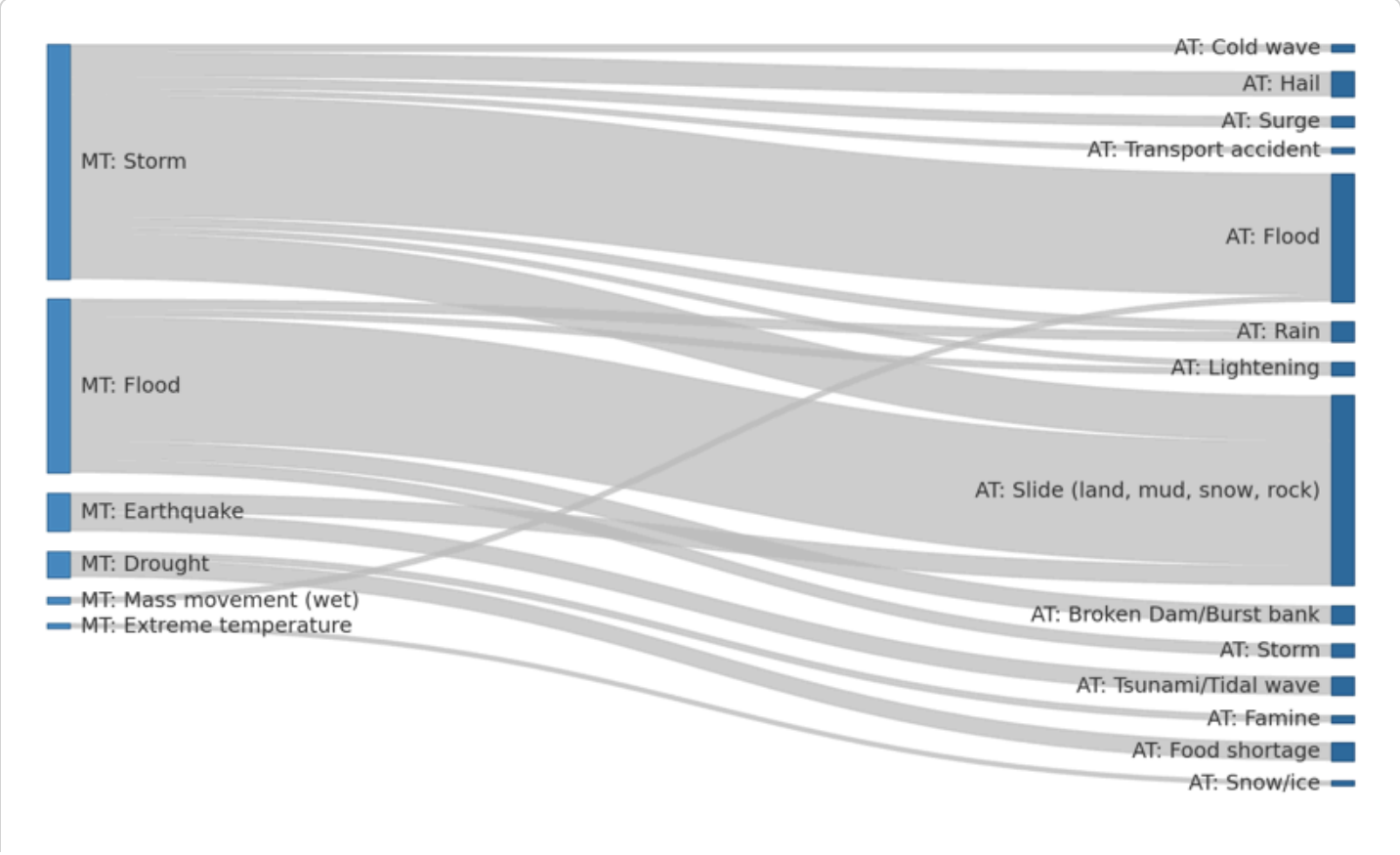
Classification Key	Group	Subgroup	Type	Subtype	Count[6]
nat-geo-mmd-sub	Natural	Geophysical	Mass movement (dry)	Sudden Subsidence (dry)	1
nat-geo-vol-ash	Natural	Geophysical	Volcanic activity	Ash fall	251
nat-geo-vol-lah	Natural	Geophysical	Volcanic activity	Lahar	1
nat-geo-vol-lav	Natural	Geophysical	Volcanic activity	Lava flow	11
nat-geo-vol-pyr	Natural	Geophysical	Volcanic activity	Pyroclastic flow	4
nat-geo-vol-vol	Natural	Geophysical	Volcanic activity	Volcanic activity (General)	11
nat-hyd-flo-coa	Natural	Hydrological	Flood	Coastal flood	86
nat-hyd-flo-fla	Natural	Hydrological	Flood	Flash flood	919
nat-hyd-flo-flo	Natural	Hydrological	Flood	Flood (General)	2366
nat-hyd-flo-ice	Natural	Hydrological	Flood	Ice jam flood	0
nat-hyd-flo-riv	Natural	Hydrological	Flood	Riverine flood	2678
nat-hyd-mmw-ava	Natural	Hydrological	Mass movement (wet)	Avalanche (wet)	122
nat-hyd-mmw-lan	Natural	Hydrological	Mass movement (wet)	Landslide (wet)	642
nat-hyd-mmw-mud	Natural	Hydrological	Mass movement (wet)	Mudslide	79
nat-hyd-mmw-roc	Natural	Hydrological	Mass movement (wet)	Rockfall (wet)	3
nat-hyd-mmw-sub	Natural	Hydrological	Mass movement (wet)	Sudden Subsidence (wet)	1
nat-hyd-wav-rog	Natural	Hydrological	Wave action	Rogue wave	0
nat-hyd-wav-sei	Natural	Hydrological	Wave action	Seiche	0
nat-met-ext-col	Natural	Meteorological	Extreme temperature	Cold wave	305
nat-met-ext-hea	Natural	Meteorological	Extreme temperature	Heat wave	344
nat-met-ext-sev	Natural	Meteorological	Extreme temperature	Severe winter conditions	79
nat-met-fog-fog	Natural	Meteorological	Fog	Fog	1

Classification Key	Group	Subgroup	Type	Subtype	Count[6]
nat-met-sto-bli	Natural	Meteorological	Storm	Blizzard/Winter storm	233
nat-met-sto-der	Natural	Meteorological	Storm	Derecho	6
nat-met-sto-ext	Natural	Meteorological	Storm	Extra-tropical storm	178
nat-met-sto-hai	Natural	Meteorological	Storm	Hail	115
nat-met-sto-lig	Natural	Meteorological	Storm	Lightning/Thunderstorms	189
nat-met-sto-san	Natural	Meteorological	Storm	Sand/Dust storm	23
nat-met-sto-sev	Natural	Meteorological	Storm	Severe weather	302
nat-met-sto-sto	Natural	Meteorological	Storm	Storm (General)	900
nat-met-sto-sur	Natural	Meteorological	Storm	Storm surge	10
nat-met-sto-tor	Natural	Meteorological	Storm	Tornado	303
nat-met-sto-tro	Natural	Meteorological	Storm	Tropical cyclone	2639
tec-ind-che-che	Technological	Industrial accident	Chemical spill	Chemical spill	106
tec-ind-col-col	Technological	Industrial accident	Collapse (Industrial)	Collapse (Industrial)	189
tec-ind-exp-exp	Technological	Industrial accident	Explosion (Industrial)	Explosion (Industrial)	790
tec-ind-fir-fir	Technological	Industrial accident	Fire (Industrial)	Fire (Industrial)	221
tec-ind-gas-gas	Technological	Industrial accident	Gas leak	Gas leak	63
tec-ind-ind-ind	Technological	Industrial accident	Industrial accident (General)	Industrial accident (General)	126
tec-ind-oil-oil	Technological	Industrial accident	Oil spill	Oil spill	8
tec-ind-poi-poi	Technological	Industrial accident	Poisoning	Poisoning	77
tec-ind-rad-rad	Technological	Industrial accident	Radiation	Radiation	9
tec-mis-col-col	Technological	Miscellaneous accident	Collapse (Miscellaneous)	Collapse (Miscellaneous)	310
tec-mis-exp-exp	Technological	Miscellaneous accident	Explosion (Miscellaneous)	Explosion (Miscellaneous)	225
tec-mis-fir-fir	Technological	Miscellaneous accident	Fire (Miscellaneous)	Fire (Miscellaneous)	821

Classification Key	Group	Subgroup	Type	Subtype	Count[6]
tec-mis-mis-mis	Technological	Miscellaneous accident	Miscellaneous accident (General)	Miscellaneous accident (General)	282
tec-tra-air-air	Technological	Transport	Air	Air	1096
tec-tra-rai-rai	Technological	Transport	Rail	Rail	647
tec-tra-roa-roa	Technological	Transport	Road	Road	2937
tec-tra-wat-wat	Technological	Transport	Water	Water	1672

Associated Disasters

In addition to the main classification system, EM-DAT makes it possible to refer to “associated disasters” to describe disaster events in more detail (see `Associated Types` in EM-DAT Public Table). They represent subsequent or co-occurring hazards that may have contributed to the disaster impact. These associated disasters may not fit into the main classification system and do not have a hierarchical structure. This additional tagging system allows for a better description of disaster events, particularly multi-hazard ones. The figure below shows the main associations found in the database.



Sankey diagram of the associations between main disaster types (MT) and associated disaster types (AT) in EM-DAT (1900–2024). The figure only reports MT-AT associations with at least 50 occurrences in the database, which account for 88% of all associations. Gray band widths are proportional to the occurrence of the association in EM-DAT. Associated types are labeled as they appear in the Associated Types column. Source: EM-DAT Archive 2.0 (April 30, 2026).

About 15% of disaster entries in EM-DAT have an associated disaster type, and only 3.5% mention two or more associated types. The most common associations are floods with landslides (23% of associations), storms with floods (22%), and storms with landslides (8%). Earthquakes are sometimes associated with landslides (4%) and tsunamis (3%) when their damage is deemed negligible compared to ground movement damage.

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1. Guha-Sapir, D., & Misson, C. (1992). The development of a database on disasters. *Disasters*, *16*(1), 74–80. https://doi.org/10.1111/j.1467-7717.1992.tb00378.x ↩

2. DHA-UNDRO, IDNDR, UNEP, WFP, WHO/PAHO, USAID/FHA, IFRC, & CRED. (1992). *Proposed principles and guidelines for the collection and dissemination of disaster-related data*. ↩

3. Below, R., Wirtz, A., & Guha-Sapir, D. (2009). *Disaster category classification and peril terminology for operational purposes*. Centre for Research on the Epidemiology of Disasters; Munich Re. ↩

4. Below, R., Grover-Kopec, E., & Dilley, M. (2007). Documenting drought-related disasters: A global reassessment. *The Journal of Environment & Development*, *16*(3), 328–344. https://doi.org/10.1177/1070496507306222 ↩

5. Integrated Research on Disaster Risk. (2014). *Peril classification and hazard glossary* (IRDR DATA Publication No. 1). Integrated Research on Disaster Risk. ↩

6. Number of disaster records at the country level in the EM-DAT Archive 2.0 (1900–2024, released April 30, 2026). ↩

5 - Classification Glossary

Definitions of Disaster Types

EM-DAT's definitions related to the group of natural hazards mainly refer to the IRDR Peril Classification and Hazard Glossary[1] (see A Brief History of the EM-DAT Classification System). These are reported in the following sections by disaster subgroups.

EM-DAT definitions related to the technological group are listed separately in the Technological Hazards section. These are legacy definitions from the EM-DAT project and do not refer to a particular glossary serving as an international standard.

> ### About Definitions
>
> Some definitions have undergone minor modifications or derivations, which are noted at the end of each section. Some terms, which are not explicitly defined in the referenced glossaries, have been left undefined. As mentioned in the Disaster Classification System, the Hazard Information Profiles (HIPs) serve as a comprehensive glossary for those interested in the definitions of different hazard types that are beyond the EM-DAT scope.

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1. Integrated Research on Disaster Risk. (2014). *Peril classification and hazard glossary* (IRDR DATA Publication No. 1). Integrated Research on Disaster Risk. ↩

5.1 - Biological Hazards

Term	Level	Definition	Source
Biological hazard	Subgroup	A hazard caused by exposure to living organisms and/or their toxic substances (e.g., venom, or mold) or vector-borne diseases that they may carry. Examples are venomous wildlife and insects, poisonous plants, algae blooms, and mosquitoes carrying agents that cause disease such as parasites, bacteria, or viruses (e.g., malaria).	IRDR
Animal incident	Type, Subtype	Human encounters with dangerous or exotic animals in both urban and rural environments.	IRDR
Epidemic	Type	Either an unusual, often sudden, increase in the number of cases of an infectious disease that already existed in the region (e.g., flu, or E. coli) or the appearance of an infectious disease previously absent from the region (e.g., plague, or polio).	EM-DAT[1]
Infectious disease	Subtype (General)	Either an unusual, often sudden, increase in the number of cases of an infectious disease that already existed in the region (e.g., flu, or E. coli) or the appearance of an infectious disease previously absent from the region (e.g., plague, or polio).	IRDR[1]
Bacterial disease	Subtype	An unusual increase in the number of cases caused by exposure to bacteria either through skin contact, ingestion, or inhalation. Examples include salmonella, methicillin-resistant *Staphylococcus aureus* (MRSA), and cholera, among others.	IRDR[2]
Parasitic disease	Subtype	An unusual increase in the number of cases caused by exposure to a parasite, i.e., an organism living on or in a host. Exposure to parasites occurs mostly through contaminated water, food, or contact with insects, animals, etc. Examples are malaria, Chagas disease, giardiasis, and trichinellosis.	IRDR[2]
Viral disease	Subtype		None[3]
Fungal disease	Subtype	An unusual increase in the number of cases caused by exposure to fungi either through skin contact, ingestion, or inhalation of spores. Examples are fungal pneumonia, fungal meningitis, etc.	IRDR[2]
Prion disease	Subtype	A type of biological hazard caused by prion proteins. Prion diseases or transmissible spongiform encephalopathies (TSEs) are a family of rare progressive neurodegenerative disorders that affect both humans and animals. They are characterized by long incubation periods and neural loss. Examples are Bovine Spongiform Encephalopathy (BSE), Creutzfeldt-Jakob Disease (CJD), Kuru, etc.	IRDR[2]
Infestation	Type, Subtype (General)	The pervasive influx, swarming and/or hatching of insects, worms, or other animals affecting humans, animals, crops, and perishable goods.	IRDR[4]
Grasshopper infestation	Subtype	Infestation of grasshoppers (insects).	IRDR[4]

Term	Level	Definition	Source
Locust infestation	Subtype	Infestation of locusts (insects).	IRDR[4]
Worms infestation	Subtype	Infestation of worms.	IRDR[4]

1. The “Infectious disease” definition corresponds to the definition of “Disease” in the IRDR glossary. The same definition is used to define “Epidemic” in the EM-DAT glossary. ↩ ↩
2. With minor modifications. ↩ ↩ ↩ ↩
3. Under the last revision of the EM-DAT classification system, certain definitions were absent from legacy or official glossaries. In such instances, the definitions have been left blank. ↩
4. These definitions have been adapted and derived from the IRDR definition of “Insect infestation”. ↩ ↩ ↩ ↩

5.2 - Climatological Hazards

Term	Level	Definition	Source
Climatological hazard	Subgroup	A hazard caused by long-lived, meso- to macro-scale atmospheric processes ranging from intra-seasonal to multi-decadal climate variability.	IRDR
Drought	Type, Subtype	An extended period of unusually low precipitation that produces a shortage of water for people, animals, and plants. Drought is different from most other hazards in that it develops slowly, sometimes over several years, and its onset is generally difficult to detect. Drought is not solely a physical phenomenon because its impacts can be exacerbated by human activities and water supply demands. Drought is therefore often defined both conceptually and operationally. Operational definitions of drought, i.e., the degree of precipitation reduction that constitutes a drought, vary by locality, climate, and environmental sector.	IRDR
Glacial lake outburst flood	Type, Subtype	These floods occur when water held back by a glacier or moraine is suddenly released. Glacial lakes can be at the front of the glacier (marginal lake) or below the ice sheet (sub-glacial lake).	IRDR[1]
Wildfire	Type, Subtype (General)	Any uncontrolled and non-prescribed combustion or burning of plants in a natural setting such as a forest, grassland, brushland or tundra, which consumes natural fuels and spreads based on environmental conditions (e.g., wind, or topography). Wildfires can be triggered by lightning or human actions.	IRDR
Forest fire	Subtype	A type of wildfire in a wooded area.	IRDR
Land fire (Brush, Bush, Pasture)	Subtype	A type of wildfire in a brush, bush, pasture, grassland, or other treeless natural environment.	IRDR[2]

1. The definition of “Glacial lake outburst flood” corresponds to the definition of “Glacial lake outburst” in the IRDR glossary. ↩

2. Not defined in the IRDR glossary but adapted from the “Wildfire” and “Forest fire” definitions. ↩

5.3 - Extra-terrestrial Hazards

Term	Level	Definition	Source
Extra-terrestrial hazard	Subgroup	A hazard caused by asteroids, meteoroids, and comets as they pass near to the Earth, enter the Earth's atmosphere, and/or strike the Earth, and by changes in interplanetary conditions that affect the Earth's magnetosphere, ionosphere, and thermosphere.	IRDR
Impact	Type	A type of extra-terrestrial hazard caused by the collision of a meteoroid, asteroid, or comet with the Earth.	IRDR
Airburst	Subtype	An explosion of a comet or meteoroid within the Earth's atmosphere without striking the ground.	IRDR
Collision	Subtype	An impact caused by the collision of a meteoroid, asteroid, or comet with the Earth's ground.	IRDR[1]
Space weather	Type	A general term for extra-terrestrial weather conditions driven by solar eruptions such as geomagnetic storms, radio disturbances, and solar energetic particles.	IRDR
Energetic particles	Subtype	Emissions from solar radiation storms consisting of pieces of matter (e.g., protons and other charged particles) moving at very high speed. The magnetosphere and atmosphere block (solar) energetic particles (SEP) from reaching humans on Earth but they are damaging to the electronics of space-borne technology (such as satellites) and pose a radiation hazard to life in space and aircraft traveling at high altitudes.	IRDR
Geomagnetic storm	Subtype	A type of extra-terrestrial hazard caused by solar wind shockwaves that temporarily disturb the Earth's magnetosphere. Geomagnetic storms can disrupt power grids, spacecraft operations, and satellite communications.	IRDR
Shockwave	Subtype	A shockwave carries energy from a disturbance through a medium (solid, liquid, or gas) similar to the action of a wave, though it travels at much higher speed. It can be a type of extra-terrestrial hazard caused by the explosion (airburst) or impact of meteorites that generate energy shockwaves capable of shattering glass, collapsing walls, etc.	IRDR
Radio disturbance	Subtype	Triggered by X-ray emissions from the Sun hitting the Earth's atmosphere and causing disturbances in the ionosphere such as jamming of high and/or low frequency radio signals. This affects satellite radio communication and Global Positioning Systems (GPS).	IRDR

1. The "Collision" definition is derived from the IRDR "Impact" and "Airburst" definitions. ↩

5.4 - Geophysical Hazards

Term	Level	Definition	Source
Geophysical hazard	Subgroup	A hazard originating from solid earth. This term is used interchangeably with the term geological hazard.	IRDR
Earthquake	Type	Sudden movement of a block of the Earth's crust along a geological fault and associated ground shaking.	IRDR
Ground movement	Subtype	Surface displacement of earthen materials due to ground shaking triggered by earthquakes or volcanic eruptions.	IRDR
Tsunami	Subtype	A series of waves (with long wavelengths when traveling across the deep ocean) that are generated by a displacement of massive amounts of water through underwater earthquakes, volcanic eruptions, or landslides. Tsunami waves travel at very high speed across the ocean, but as they begin to reach shallow water they slow down, and the wave grows steeper.	IRDR
Mass movement (dry)	Type	Any type of downslope movement of earth materials under hydrological dry conditions.	IRDR[1]
Avalanche (dry)	Subtype	A large mass of loosened earth material, snow, or ice that slides, flows, or falls rapidly down a mountainside under the force of gravity. Debris Avalanche: The sudden and very rapid downslope movement of a mixed mass of rock and soil. There are two general types of debris avalanches. A cold debris avalanche usually results from an unstable slope suddenly collapsing whereas a hot debris avalanche results from volcanic activity leading to slope instability and collapse.	IRDR
Landslide (dry)	Subtype	Any kind of moderate to rapid soil movement incl. lahars, mudslides, and debris flows (under dry conditions). A landslide is the movement of soil or rock controlled by gravity and the speed of the movement usually ranges between slow and rapid, but it is not very slow. It can be superficial or deep, but the materials must make up a mass that is a portion of the slope or the slope itself. The movement has to be downward and outward with a free face.	EM-DAT
Rockfall (dry)	Subtype		None[2]
Sudden subsidence (dry)	Subtype	Sinking of the ground due to groundwater removal, mining, dissolution of limestone (e.g., karst sinkholes), extraction of natural gas, and earthquakes. In this case, the sinking occurs under dry conditions as a result of a geophysical trigger.	IRDR[3]
Volcanic activity	Type, Subtype (General)	A type of volcanic event near an opening/vent in the Earth's surface including volcanic eruptions of lava, ash, hot vapor, gas, and pyroclastic material.	IRDR
Ash fall	Subtype	Fine (less than 4 mm in diameter) unconsolidated volcanic debris blown into the atmosphere during an eruption; can remain airborne for long periods of time and travel a considerable distance from the source.	IRDR

Term	Level	Definition	Source
Lava flow	Subtype	The ejected magma that moves as a liquid mass downslope from a volcano during an eruption.	IRDR
Pyroclastic flow	Subtype	Extremely hot gases, ash, and other materials with a temperature of more than 1,000 degrees Celsius that rapidly flow down the flank of a volcano (at more than 700 km/h) during an eruption.	IRDR
Lahar	Subtype	Hot or cold mixture of earthen material flowing down the slope of a volcano either during or between volcanic eruptions.	IRDR

1. The definition of “Mass movement (dry)” is adapted from the “Mass movement” IRDR definition. ↩
2. Under the last revision of the EM-DAT classification system, certain definitions were absent from legacy or official glossaries. In such instances, the definitions have been left blank. ↩
3. The first definition sentence of “Sudden subsidence (dry)” is the definition of “Subsidence” in the IRDR glossary. The second sentence has been added to distinguish this class from “Sudden subsidence (wet)” in the hydrological group. ↩

5.5 - Hydrological Hazards

Term	Level	Definition	Source
Hydrological hazard	Subgroup	A hazard caused by the occurrence, movement, and distribution of surface and subsurface freshwater and saltwater.	IRDR
Flood	Type, Subtype (General)	A general term for the overflow of water from a stream channel onto normally dry land in the floodplain (riverine flooding), higher-than-normal levels along the coast (coastal flooding) and in lakes or reservoirs as well as ponding of water at or near the point where the rain fell (flash floods).	IRDR
Coastal flood	Subtype	Higher-than-normal water levels along the coast caused by tidal changes or thunderstorms that result in flooding, which can last from days to weeks.	IRDR
Flash flood	Subtype	Heavy or excessive rainfall in a short period of time that produces immediate runoff, creating flooding conditions within minutes or a few hours during or after the rainfall.	IRDR
Riverine flood	Subtype	A type of flooding resulting from the overflow of water from a stream or river channel onto normally dry land in the floodplain adjacent to the channel.	IRDR
Ice jam flood	Subtype	The accumulation of floating ice restricting or blocking a river's flow and drainage. Ice jams tend to develop near river bends and obstructions (e.g., bridges).	IRDR
Mass movement (wet)	Type	Types of mass movement that occur when heavy rain or rapid snow/ice melt send large amounts of vegetation, mud, or rock down a slope driven by gravitational forces.	IRDR[1]
Avalanche (wet)	Subtype	A large mass of loosened earth material, snow, or ice that slides, flows, or falls rapidly down a mountainside under the force of gravity. Snow Avalanche: Rapid downslope movement of a mix of snow and ice.	IRDR
Landslide (wet)	Subtype	Any kind of moderate to rapid soil movement incl. lahars, mudslides, and debris flows (under wet conditions). A landslide is the movement of soil or rock controlled by gravity and the speed of the movement usually ranges between slow and rapid, but it is not very slow. It can be superficial or deep, but the materials must make up a mass that is a portion of the slope or the slope itself. The movement has to be downward and outward with a free face.	EM-DAT
Rockfall (wet)	Subtype		None[2]
Sudden subsidence (wet)	Subtype	Sinking of the ground due to groundwater removal, mining, dissolution of limestone (e.g., karst sinkholes), extraction of natural gas, and earthquakes. In this case, the sinking occurs under wet conditions as a result of a hydrological trigger (e.g., rain).	IRDR[3]
Mudslide	Subtype		None[2]
Wave action	Type	Wind-generated surface waves that can occur on the surface of any open body of water such as oceans, rivers, or lakes. The size of the wave depends on the	IRDR

Term	Level	Definition	Source
		strength of the wind and the distance traveled (fetch).	
Rogue wave	Subtype	An unusual single crest of an ocean wave far out at sea that is much higher and/or steeper than other waves in the prevailing swell system.	IRDR
Seiche	Subtype	A standing wave of water in a large semi- or fully-enclosed body of water (lakes or bays) created by strong winds and/or a large barometric pressure gradient.	IRDR

1. The “Mass movement (wet)” definition is adapted from the IRDR definition of “Debris flow, mud flow, rock fall”. ↩
2. Under the last revision of the EM-DAT classification system, certain definitions were absent from legacy or official glossaries. In such instances, the definitions have been left blank. ↩ ↩
3. The first definition sentence of “Sudden subsidence (wet)” is the definition of “Subsidence” in the IRDR glossary. The second sentence has been added to distinguish this class from “Sudden subsidence (dry)” in the geophysical group. ↩

5.6 - Meteorological Hazards

Term	Level	Definition	Source
Meteorological hazard	Subgroup	A hazard caused by short-lived, micro- to meso-scale extreme weather and atmospheric conditions that last from minutes to days.	IRDR
Extreme temperature	Type	A general term for temperature variations above (extreme heat) or below (extreme cold) normal conditions.	IRDR
Cold wave	Subtype	A period of abnormally cold weather. Typically, a cold wave lasts for two or more days and may be aggravated by high winds. The exact temperature criteria for what constitutes a cold wave may vary by location.	EM-DAT
Heat wave	Subtype	A period of abnormally hot and/or unusually humid weather. Typically, a heat wave lasts for two or more days. The exact temperature criteria for what constitutes a heat wave may vary by location.	EM-DAT
Severe winter conditions	Subtype	Damage caused by snow and ice. Winter damage refers to damage to buildings, infrastructure, traffic (esp. navigation) inflicted by snow and ice in the form of snow pressure, freezing rain, frozen waterways etc.	EM-DAT
Fog	Type, Subtype	Water droplets that are suspended in the air near the Earth's surface. Fog is, in fact, simply a cloud that is in contact with the ground.	IRDR[1]
Storm	Type, Subtype (General)		None[2]
Derecho	Subtype	Widespread and usually fast-moving windstorms associated with a convection/convective storm. Derechos include downburst and straight-line winds. The damage from derechos is often confused with the damage from tornadoes.	IRDR
Hail	Subtype	Solid precipitation in the form of irregular pellets or balls of ice more than 5 mm in diameter.	IRDR
Lightning/Thunderstorms	Subtype	A high-voltage, visible electrical discharge produced by a thunderstorm and followed by the sound of thunder.	IRDR[3]
Sand/Dust storm	Subtype	Strong winds carrying particles of sand aloft, but generally confined to less than 50 feet (15 m), especially common in arid and semi-arid environments. A dust storm is also characterized by strong winds but carries smaller particles of dust rather than sand over an extensive area.	IRDR
Storm surge	Subtype	An abnormal rise in sea level generated by a tropical cyclone or other intense types of storm.	IRDR

Term	Level	Definition	Source
Tornado	Subtype	A violently rotating column of air that reaches the ground or open water (waterspout).	IRDR
Winter storm/Blizzard	Subtype	A low-pressure system in winter months with significant accumulations of snow, freezing rain, sleet, or ice. A blizzard is a severe snowstorm with winds exceeding 35 mph (56 km/h) for three or more hours, producing reduced visibility (less than 0.25 miles (400 m)).	IRDR
Extra-tropical storm	Subtype	A type of low-pressure cyclonic system in the middle and high latitudes (also called a mid-latitude cyclone) that primarily gets its energy from the horizontal temperature contrasts (fronts) in the atmosphere. When associated with cold fronts, extra-tropical cyclones may be particularly damaging (e.g., European winter/windstorm, or Nor'easter).	IRDR
Tropical cyclone	Subtype	A tropical cyclone originates over tropical or subtropical waters. It is characterized by a warm-core, non-frontal synoptic-scale cyclone with a low-pressure center, spiral rain bands and strong winds. Depending on their location, tropical cyclones are referred to as hurricanes (Atlantic, Northeast Pacific), typhoons (Northwest Pacific), or cyclones (South Pacific and Indian Ocean).	IRDR
Severe weather	Subtype		None[2]

1. Note: the only "Fog" entry in EM-DAT is the Great London SMOG, 1952, which was also accompanied by air pollution. ↩
2. Under the last revision of the EM-DAT classification system, certain definitions were absent from legacy or official glossaries. In such instances, the definitions have been left blank. ↩ ↩
3. The "Lightning/Thunderstorms" definition corresponds to the definition of "Lightning" in the IRDR glossary. ↩

5.7 - Technological Hazards

Term	Level	Definition	Source
Industrial accident	Subgroup, Type, Subtype (General)	Disaster type term used in EM-DAT to describe technological accidents of an industrial nature/involving industrial buildings (e.g., factories).	EM-DAT
Miscellaneous accident	Subgroup, Type, Subtype (General)	Disaster type term used in EM-DAT to describe technological accidents of a non-industrial or transport nature (e.g., involving houses).	EM-DAT
Chemical spill	Type, Subtype	Accidental release occurring during the production, transportation, or handling of hazardous chemical substances.	EM-DAT
Collapse (Industrial / Miscellaneous)	Type, Subtype	Accident involving the collapse of a building or structure. Can either involve industrial structures or domestic/non-industrial structures.	EM-DAT
Explosion (Industrial / Miscellaneous)	Type, Subtype	Explosions involving buildings or structures. Can involve industrial structures.	EM-DAT
Fire (Industrial / Miscellaneous)	Type, Subtype	Urban fire involving buildings or structures. Can involve industrial structures.	EM-DAT[1]
Gas leak	Type, Subtype		None[2]
Oil spill	Type, Subtype		None[2]
Poisoning	Type, Subtype	Poisoning of atmosphere or water courses due to industrial sources of contamination.	EM-DAT
Radiation	Type, Subtype		None[2]
Transport accident	Subgroup	Disaster type term used to describe technological transport accidents involving mechanized modes of transport. It comprises four disaster types (i.e., Air, Water, Rail, and Road).	EM-DAT
Air	Type, Subtype	Transport accidents involving airplanes, helicopters, airships, and balloons.	EM-DAT
Water	Type, Subtype	Transport accidents involving sailing boats, ferries, cruise ships, and other vessels.	EM-DAT
Rail	Type, Subtype	Transport accidents involving trains.	EM-DAT
Road	Type, Subtype	Transport accidents involving motor vehicles on roads and tracks.	EM-DAT

1. Urban fire is a former class that is now either described as “Fire (Industrial)” or “Fire (Miscellaneous)”. ↩

2. Under the last revision of the EM-DAT classification system, certain definitions were absent from legacy or official glossaries. In such instances, the definitions have been left blank. ↩ ↩ ↩

6 - Hazard and Disaster Magnitude Units

Additional Information About the Hazard Extent

Some disaster types may have a reported magnitude in the `Magnitude` and `Magnitude Scale` columns of the EM-DAT Public Table. EM-DAT disaster magnitude scales vary depending on the disaster type. The table below specifies the magnitude property, unit, and exact value found in the `Magnitude Scale` column for the related disaster types. Note that the `Magnitude Scale` column is filled for all disasters of these types, even when no magnitude value is reported.

Disaster Type	Magnitude property	Magnitude Unit or Scale	Value in `Magnitude Scale`
Earthquake	Size of the earthquake derived from its physical properties	Moment Magnitude[1]	`Moment Magnitude`
Flood	Flood extent (area)	km^2 (square kilometers)	`Km2`
Drought	Drought extent (area)	km^2 (square kilometers)	`Km2`
Extreme temperature	The recorded extreme temperature (maximum or minimum depending on whether it is a heat or a cold wave)	°C	`°C`
Epidemic	Number of vaccinated people	Vaccinated people[2]	`Vaccinated`
Wildfire	Wildfire extent (area)	km^2 (square kilometers)	`Km2`
Storm	Recorded wind speed	km/h (kilometers per hour)	`Kph`
Industrial accident (all types)[3]	Chemical discharged volume	m^3 (cubic meters)	`m3`

1. See the following USGS Document. ↩
2. As the magnitude column provides additional hazard-specific information, it is used to report the number of vaccinated people for epidemics. However, it may not be a good indicator of the magnitude of the epidemic, which is also captured by the human impact. ↩
3. The `m3` scale is set for all industrial accident types, but magnitude values are reported almost exclusively for chemical spills. ↩

7 - Impact Variables

What Type of Information Is Collected to Evaluate the Impact of a Disaster? How Is It Aggregated, Reported, or Adjusted?

7.1 - Human Impact Variables

People Affected and Death Toll

Five variables describe the human impact of disasters in the EM-DAT Public Table:

> ### Human Impact Variables
>
> 1. `Total Deaths`, which includes reported deaths and missing people,
> 2. `No. Injured`,
> 3. `No. Affected`,
> 4. `No. Homeless`, and
> 5. `Total Affected`, which is the sum of `No. Injured`, `No. Affected`, and `No. Homeless`.

The reported total number of deaths (column `Total Deaths`) includes confirmed fatalities directly attributed to the disaster plus missing people whose whereabouts since the disaster are unknown and so they are presumed dead based on official figures.

Aside from fatalities, the number of injured people (column `No. Injured`) is entered when the term “injured” is written in the source. Any related word like “hospitalized” is considered as injured. If no precise number is given, such as “hundreds of injured,” 200 injured will be entered (although this figure is probably an underestimate).

The number of affected people (column `No. Affected`) is often mentioned and is widely used by different actors to convey the extent, impact, or severity of a disaster in non-spatial terms. If only the number of families affected or houses damaged are reported, the figure is multiplied by the average family size for the affected area (×5 for developing countries, ×3 for industrialized countries, according to the UNDP country classification).

> ### Calculation of `No. Affected` (Examples)

- Number of houses damaged: 50 × 5 = 250 affected (although this figure is probably an underestimate).
- If the value ranges from a minimum to a maximum: the average is taken.
- Thousands affected: 2,000 affected (although this figure is probably an underestimate).

Similarly, the indicator `No. Homeless` is mentioned whenever it is found in reports. If only the number of families that are homeless or houses that are destroyed are reported, the figure is multiplied by the average family size for the affected area (×5 for developing countries, ×3 for industrialized countries, according to the UNDP country classification).

Calculation of `No. Homeless` (Examples)

- Number of houses destroyed: 50 × 5 = 250 homeless (although this figure is probably an underestimate).
- If the value ranges from a minimum to a maximum: the average is taken.
- Thousands homeless: 2,000 homeless (although this figure is probably an underestimate).

Finally, the indicator `Total Affected` is the total of injured, affected, and homeless people. For each disaster and its corresponding sources, the population referred to in these statistics and the apportionment between injured, affected, homeless, and the total is carefully checked by CRED staff members.

Human Impact and Conceptual Uncertainties

The terms “dead,” “injured,” and “homeless” are more straightforward than “affected persons.” The definition of “affected persons” varies across sources and lacks a universal standard. Besides, for disasters such as droughts or heat waves, which have ambiguous spatiotemporal boundaries, determining the number of casualties can be challenging. CRED directly uses numbers in EM-DAT as they appear in original sources, even if there are uncertainties in these figures. More details on these uncertainties can be found in the section on General Issues.

Usually, at least the field `Total Deaths` or `Total Affected` is found in EM-DAT records as these numbers are involved as entry criteria. However, records often contain incomplete impact statistics (see Accounting Biases).

7.2 - Economic Impact Variables

Total Economic Damage, Reconstruction Costs, and Insured Damage

Six variables describe the economic impact of disasters in the EM-DAT Public Table:

Economic Impact Variables

1. `Reconstruction Costs ('000 US$)`,
2. `Reconstruction Costs, Adjusted ('000 US$)`,
3. `Insured Damage ('000 US$)`,
4. `Insured Damage, Adjusted ('000 US$)`,
5. `Total Damage ('000 US$)`, and
6. `Total Damage, Adjusted ('000 US$)`.

These six statistics are the same three statistics (`Reconstruction Costs ('000 US$)`, `Insured Damage ('000 US$)`, and `Total Damage ('000 US$)`) repeated to also provide an amount corrected for inflation, i.e., "Adjusted" (see Economic Adjustment). Damage and costs are converted and expressed in thousands of US dollars ('000 US$).

Underreporting of Economic Losses

Economic damage resulting from disasters is largely underreported. In EM-DAT, figures tend to be available only for high-impact disasters in countries with insurance and reinsurance coverage. This issue is described in the Specific Biases section, in particular, in the Accounting Biases section.

Reconstruction costs are different from total damage as they must consider the current construction or purchase costs of goods, as well as the additional cost of prevention and mitigation measures designed to reduce damage from future disasters. Hence, when reconstruction costs are specified, they are usually greater than the total damage.

Insured damage is usually reported by reinsurance companies that publish figures about disaster losses, e.g., Munich Re, Swiss Re, or Aon. When insured damage is reported, the total damage is generally reported from the same source for consistency.

8 - Spatial Information and Geocoding

Standards for Administrative Regions

Country Codes (ISO-3)

In the EM-DAT Public Table, the `ISO` column indicates a 3-letter (alpha-3) code representing a specific country, e.g., “BEL” for Belgium. This code is presented according to the international standard ISO 3166 determined by the International Organization for Standardization (ISO). Due to historical changes in the countries’ denomination or boundaries, you may find country codes that are not found in the current ISO 3166 alpha-3 country codes. These extensions are listed in the table below.

Country codes are particularly useful to link the EM-DAT tabular data to a spatial layer using a Geographic Information System (GIS), spatial database, or geoprocessing programming library.

United Nations M49 Standard Country or Area Codes

The UN M49, also known as the Standard Country or Area Codes for Statistical Use (Series M, No. 49), is a set of area codes formulated by the United Nations for statistical use. This standard is curated and upheld by the United Nations Statistics Division. The UN M49 alpha-3 codes largely overlap with the ISO 3166 alpha-3 codes.

Since September 2023, the EM-DAT Public Table refers to the `Country`, `Region`, and `Subregion` names as found in the UN M49 standard. Codes used for the EM-DAT extensions to the ISO alpha-3 codes and UN M49 are listed in the table below.

ISO / UN M49 Extensions

> ### Disclaimer
>
> The extensions to the ISO 3166-1 alpha-3 norm are the results of historical changes in the norms used. Please note that CRED aims to deliver data within an accurate spatiotemporal context. The way the EM-DAT data is presented and documented on this website does not reflect any geopolitical views held by CRED, or the views of our partners.

Alpha-3 Code	Country Name	Region	Subregion
ANT	Netherlands Antilles	Americas	Latin America and the Caribbean
AZO	Azores Islands	Europe	Southern Europe
CHA	Channel Islands	Europe	Western Europe
CSK	Czechoslovakia	Europe	Eastern Europe
DDR	German Democratic Republic	Europe	Western Europe
DFR	Germany Federal Republic	Europe	Western Europe
SCG	Serbia Montenegro	Europe	Southern Europe
SPI	Canary Islands	Africa	Northern Africa
SUN	Soviet Union	Europe	Eastern Europe
YMD	People's Democratic Republic of Yemen	Asia	Western Asia
YMN	Yemen Arab Republic	Asia	Western Asia
YUG	Yugoslavia	Europe	Southern Europe
TWN	Taiwan (Province of China)	Asia	Eastern Asia

GAUL 2015 Administrative Units

> ## GAUL 2015 Maintenance
>
> Maintenance of GAUL 2015 in EM-DAT ceased in 2025. Users are referred to the GADM 4.1 Administrative Units (next section) as the recommended alternative. GAUL 2015 codes are expected to be fully retired by the end of 2027.

Between 2014 and 2025, EM-DAT relied on the Global Administrative Unit Layers (GAUL 2015) implemented by the Food and Agriculture Organization (FAO) to indicate in the `Admin Units` column of the EM-DAT Public Table which region is affected by the disaster, up to the Admin-2 level (e.g., districts). The mapping of EM-DAT disaster events at this higher level of geographical precision has only been completed for non-biological natural-hazard data since 2000.

The human and economic impact variables at the country level (Admin-0) are not disaggregated between regions at the Admin-1 or Admin-2 level. Hence, only the occurrence is available at a more

precise administrative level, and the impact variables remain representative of the country level.

The geometries for GAUL 2015 are available as a Geopackage file.

GADM 4.1 Administrative Units

The GADM Database of Global Administrative Areas was selected to replace GAUL 2015 geocoded administrative units, from 2026 onwards. As with GAUL 2015 Administrative Units, geocoding remains limited to non-biological natural-hazard data since 2000. The primary geocoding workflow remains manual. GADM 4.1 geocoded units are assigned manually by database managers, using existing geocoding services, based on the `Location` column (see Public Table). However, as of January 2026, most GADM administrative units in EM-DAT for the 2000–2025 period are the result of a migration from the GAUL 2015 Administrative Units, and this migration is hereby documented to reflect potential precision losses inherent in the process.

GAUL 2015 to GADM 4.1 Migration

The complete procedure followed two steps:

1. The generation of a translation layer from GAUL to GADM. Based on geoprocessing, GAUL and GADM geometries were compared using geometric and topological indicators. Several strategies and outcomes are considered in the translation layer.
2. The optimal translation selection workflow. Among candidate GAUL-to-GADM translation options, the workflow follows a decision tree based on indicators associated with the translation layer, ultimately selecting GADM units.

Technical aspects

For more details on the technical aspects, we refer to:

- CRED Crunch No. 77 (December 2024).*
- Delforge, D., Below, R., Wathelet, V., Alonso, A., & Speybroeck, N. (2026). *The 2024 Emergency Events Database Archive: The year in review and historical data updates* [Preprint]. Zenodo. https://doi.org/10.5281/zenodo.20071134

*Since the release of CRED Crunch No. 77, additional indicators and workflows have been introduced. They are described in the methods and table below.

Translation Methods and Strategies

Candidate translation options are identified using four methods, listed below in order of priority. The companion table provides an additional summary, including the labels that could be found in the `GADM Admin Units` column (see Public Table) to identify the matching method.

1. One-to-one matching using overlap indices, i.e., the Jaccard Index (JI). Labels `jaccard1` and `jaccard2` refer to units that have been matched using this method, either at the Admin-1 or Admin-2 level.

2. One-to-many matching using a selection strategy to reconstruct a composite GADM administrative footprint matching the original GAUL one with a list of GADM units. The quality of the composite "puzzle" footprint is also evaluated using the Jaccard Index (JI). The approach can be applied at the same administrative level (`puzzle1_1` or `puzzle2_2`) or between GAUL Admin-1 and GADM Admin-2 (`puzzle1_2`). *

3. One-to-one matching using the discrete Hausdorff distance as an alternative indicator, primarily focused on the spatial proximity of points defined in nearby geometries. A buffer distance is used to identify nearby geometries. In contrast, JI is focused on overlap ratios. The routine is either applied at the Admin-1 (`bufhaus1`) or Admin-2 level (`bufhaus2`).

4. An upward fallback strategy, which consists of identifying a GADM unit containing the GAUL original one. The `parent2_1` workflow selects a `jaccard1` match for the GAUL parent unit when applicable. The `within2_1` workflow identifies the upward GADM Admin-1 unit based on the area ratio of the contained GAUL Admin-2 unit.

* While we do not provide a `jaccard1_2` label, a `puzzle1_2` method resulting in one admin unit being identified is conceptually and methodologically equivalent.

Method	Labels	Advantages	Issues
Direct Jaccard (1-1)	`jaccard1` , `jaccard2`	Simple and interpretable	Limited matching capabilities
Puzzle composition (1-n)	`puzzle1_1` , `puzzle1_2` , `puzzle2_2`	Effective in case of administrative reconfiguration	Less interpretable
Buffer + Hausdorff distance (1-1)	`bufhaus1` , `bufhaus2`	Effective in the case of small, isolated, or island territories	Computationally intensive
Upward fallback (1-1)	`parent2_1` , `within2_1`	Increases matching rates when other methods fail	Loss of precision

As of January 2026, around 90% of the GAUL 2015 footprints have been matched to corresponding GADM 4.1 footprints, satisfying quality thresholds. The remaining 10% will be geocoded progressively,

using the manual workflow.

See also: Additional Resources/Augmented EM-DAT Data

9 - EM-DAT Sources of Information

Where Do the EM-DAT Figures Come From?

The EM-DAT database is compiled from various sources, including the United Nations, governmental and non-governmental agencies, insurance companies, research institutes, and the press. The table below lists the most common sources as of September 2023. For further inquiries on the data collection and selection process, see EM-DAT Protocols.

Sources are not displayed in the public records for legal reasons, to preserve the integrity of the services EM-DAT depends on, and to uphold neutrality, especially for politically sensitive data. For a quantitative overview of the sources used between 2000 and 2023, including the top 20 sources, their documentation routines, and how often each contributes to `Total Deaths`, `Total Affected`, and `Total Damage`, see Delforge et al. (2025)[1]. Note that the set of sources evolves over time: some systematic sources have been discontinued, which may introduce biases (see Known Issues and Limitations).

Source name	Category	Type of disasters covered
Office for the Coordination of Humanitarian Affairs (OCHA) / ReliefWeb	United Nations	Natural hazards
World Food Programme (WFP)[2]	United Nations	Drought/Famine
World Meteorological Organization (WMO)	United Nations	Natural hazards
World Health Organization (WHO)	United Nations	Epidemics
Food and Agriculture Organization (FAO)[2]	United Nations	Drought/Famine
United Nations Environment Programme (UNEP)	United Nations	Natural hazards
National governments (reports)	National government	Natural and technological hazards
Federal Emergency Management Agency (FEMA)	United States	Natural hazards (United States)
National Oceanic and Atmospheric Administration (NOAA)	United States	Natural hazards
USAID Bureau for Humanitarian Assistance (BHA, formerly OFDA)[2]	United States	Natural and technological hazards
US Geological Survey (USGS)	United States	Earthquakes
Centers for Disease Control and Prevention (CDC)	United States	Epidemics

Source name	Category	Type of disasters covered
National Centers for Environmental Information (NCEI, formerly the National Geophysical Data Center, NGDC)[2]	United States	Natural hazards
European Civil Protection and Humanitarian Aid Operations (ECHO)	European Union	Natural hazards
Dartmouth Flood Observatory (DFO)	Research center	Floods, landslides, and storms
International Federation of Red Cross and Red Crescent Societies (IFRC)	Humanitarian organization	Natural and technological hazards
World Bank (reports)	Intergovernmental organization	Major natural hazards and disasters
Swiss Re	Reinsurance company	Natural and technological hazards
Munich Re	Reinsurance company	Natural hazards
Aon (formerly Aon Benfield)	Reinsurance company	Natural hazards
Lloyd's Casualty Magazine (paid subscription)[2]	Reinsurance company	Natural hazards
Agence France-Presse (paid subscription)	Press and other media	Natural and technological hazards
Reuters[2]	Press and other media	Natural and technological hazards
The New Humanitarian (formerly IRIN News)[2]	Press and other media	Natural and technological hazards (mostly Africa)
FloodList	Press and other media	Floods
Wikipedia	Press and other media	Natural and technological hazards
Plane Crash Info	Press and other media	Transport accidents (Air)

1. Delforge, D., Wathelet, V., Below, R., Lanfredi Sofia, C., Tonnelier, M., van Loenhout, J. A. F., & Speybroeck, N. (2025). EM-DAT: The Emergency Events Database. *International Journal of Disaster Risk Reduction*, *124*, Article 105509. https://doi.org/10.1016/j.ijdrr.2025.105509 ↩

2. Historical source no longer used ↩ ↩ ↩ ↩ ↩ ↩ ↩